



CHRIST

(DEEMED TO BE UNIVERSITY)

BANGALORE | DELHI NCR | PUNE

Centre for Quantum Technologies and Complex Systems

Established on 2 July, 2025

CHRIST(Deemed to be University)



Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment



Vision

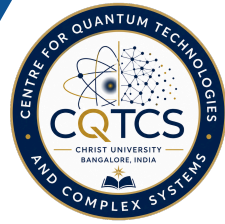
Excellence and Service



Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

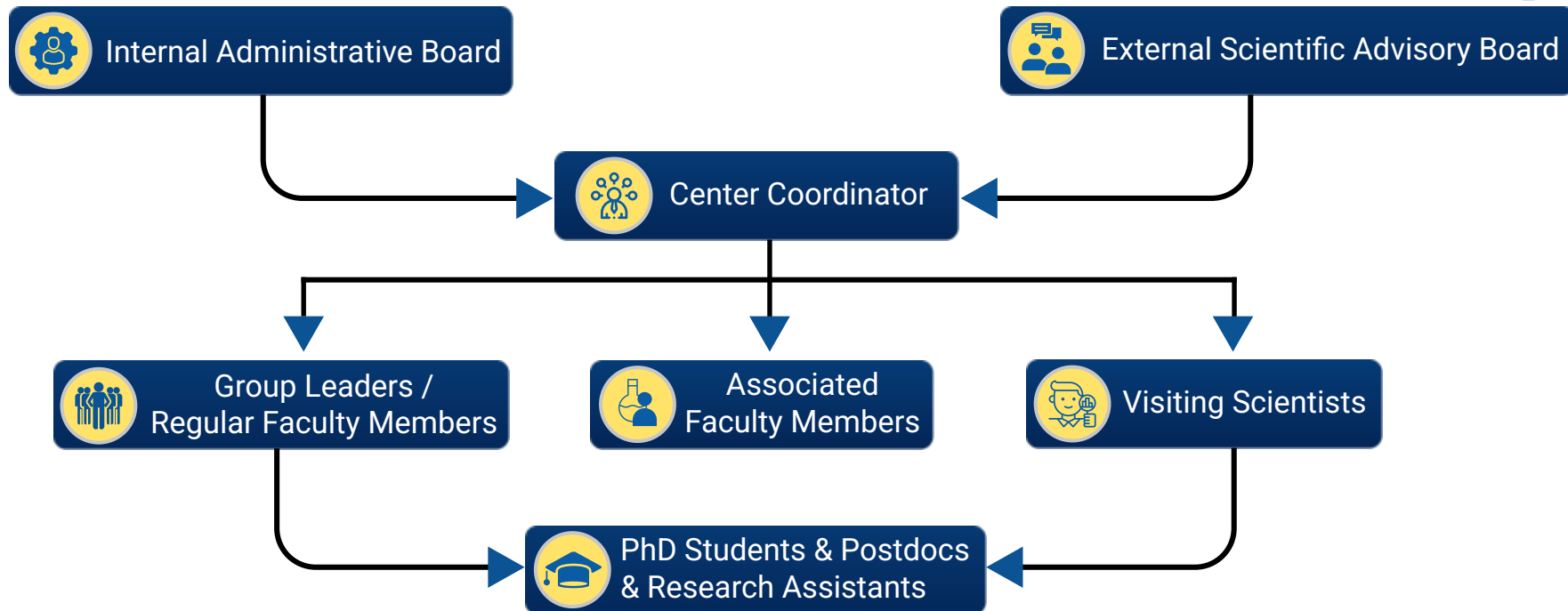
CENTER FOR QUANTUM TECHNOLOGIES AND COMPLEX SYSTEMS (CQTCS)



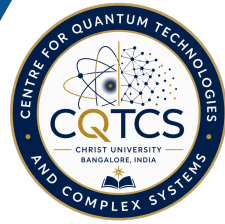
Goal of the centre

- ☐ CQTCS aims to establish an international level world-leading research centre dedicated to advancing foundational and applied research in quantum science, engineering, and complex systems dynamics.
- ☐ Centre will train the next generation of physicists and interdisciplinary researchers, and provide support for innovation in quantum technologies and complex systems
- ☐ Centre will foster international collaborations and partnerships.
- ☐ This centre follows the pattern of the world class institution Max Planck, Germany

CENTER FOR QUANTUM TECHNOLOGIES AND COMPLEX SYSTEMS (CQTCS)



CENTER FOR QUANTUM TECHNOLOGIES AND COMPLEX SYSTEMS (CQTCS)



Scientific Advisory Board



Prof. Alan R.
Bishop

Los Alamos
National LAB,



Prof. Sergej
Flach

PCS-IBS, South
Korea



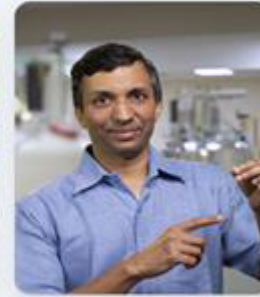
Prof. Adolfo del
Campo

Uni. Luxembourg,
Luxembourg



Prof. Arul
Lakshminarayan

IIT Madras, India



Prof. T S
Mahesh

IISER Pune, India



Prof. Pratap
Raychaudhuri

TIFR Mumbai,
India

MEET OUR GROUP LEADERS



Dr. Mithun Thudiyangal

Group Leader



Research Focus

Ultracold Quantum Matter
and Quantum Algorithms



Dr. Basudeb Sain

Group Leader



Research Focus

Photonics / Quantum
Photonics & Optical
Metasurfaces



Dr. Arindam Mallick

Group Leader



Research Focus

Quantum Information,
Quantum Simulation &
Mathematical Physics



Dr. Sreeja C S

Group Leader



Research Focus

Quantum-Safe
Cryptography
& Cyber Security



Dr. Ranjith Kumar K M

Group Leader



Research Focus

Quantum Materials &
Quantum Magnetism
Low Temperature Physics

Research Domains of CQTCS

Theory

- ☐ Dr. Mithun Thudiyangal
- ☐ Dr. Sreeja CS
- ☐ Dr. Arindam Mallick



- ☐ Quantum algorithms and Simulations
- ☐ Quantum computing and Quantum communications
- ☐ Complex Svstems

Experiment

- ☐ Dr. Basudeb Sain
- ☐ Dr. Ranjith Kumar K M



- ☐ Photonics quantum systems
- ☐ Low Temperature Physics
- ☐ Quantum Magnetism & Materials

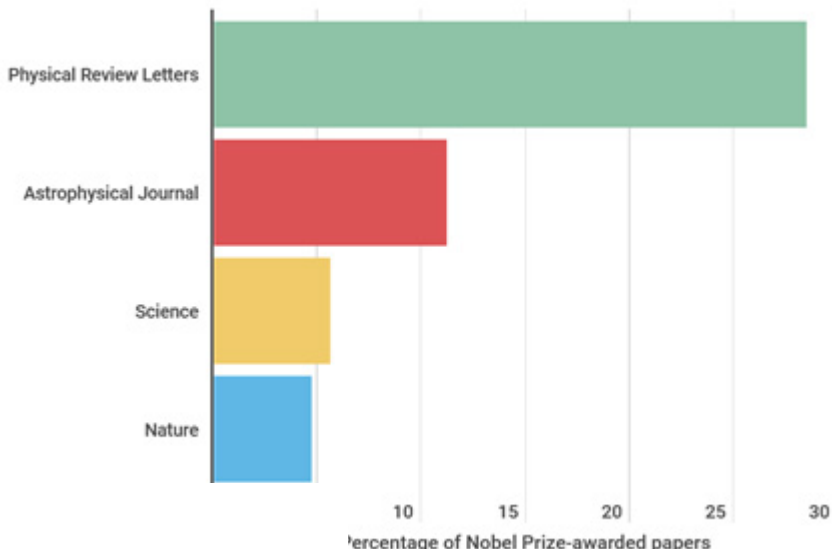
Features of Group Leaders

In interdisciplinary Journals

- ☐ 4 -Physical Review Letters -American Physical Society (APS)
- ☐ 4 -Light: Science & Applications -Nature
- ☐ 3 -Science advances -Science

In subject Journals

- ☐ > 50 APS papers
- ☐ Publications in ACS journals,
- ☐ Nano Letters, RSC Advances,
- ☐ Nanoscale, Nature Photonics, Nanophotonics etc.



Nobel Prize-winning Research in Physical Review Letters

APS's Physical Review journals are home to the most Nobel-winning physics papers in the world.

APS's Physical Review Letters publishes prize-winning research. Over 65% of the Nobel Prize-winning research published in the last four decades is included in Physical Review journals, as are Nobel Prize in Physics winners from the previous 13 years.

Collaborations

Academic

- ✓ University of Maryland, USA
- ☐ University of Luxembourg, Luxembourg
- ☐ National Institute of Standards and Technology (NIST), USA

- ☐ Institute for Basic Science, South Korea
- ☐ Lancaster University, England
- ☐ IIT Delhi, IIT Kharagpur

- ✓ Los Alamos National Laboratory, USA
- ✓ Kindai University, Japan
- ✓ University of Massachusetts, Amherst, USA
- ☐ MaxPlanck Institute, Germany
- ☐ National High magnetic field Laboratory, Grenoble, France
- ☐ Leibniz Institute (IFW) Dresden, Germany
- ☐ High Magnetic field Laboratory, Germany

Industry

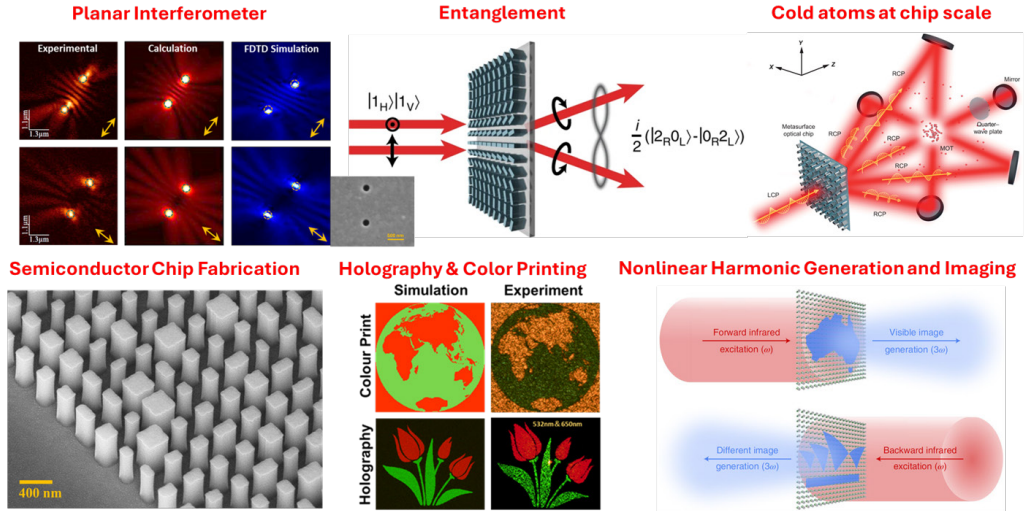
- ☐ Qutools, Germany
- ☐ Kipu Quantum, Germany
- ☐ Ansys, India
- ☐ Pernishq Delhi

MoU

- ☐ Qutools, Germany
- ☐ Ansys, India
- ☐ Space Applications Centre (SAC), ISRO

Photonics Lab Facilities

- Supercontinuum Laser System
- Chemical Nanofabrication Lab
- Photonic characterization facilities (Microscopic and Spectroscopic)



Quantum Magnetism & Materials Research



Quantum Spin Liquid (QSL)

Emergent disorder and fractionalised excitations



Spin Nematicity

Quadrupolar order and hidden phases



BEC of Triplons

Bose-Einstein condensation of magnetic excitations

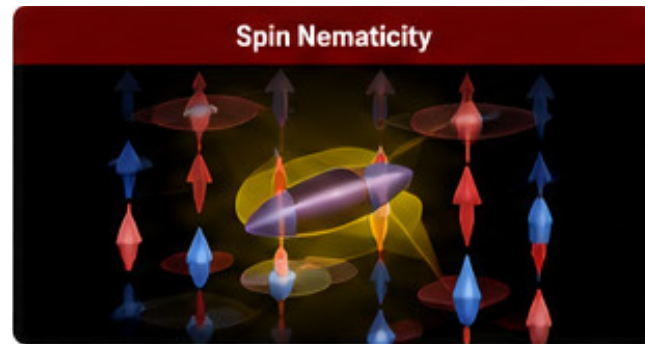


Adiabatic Cooling (ADR)

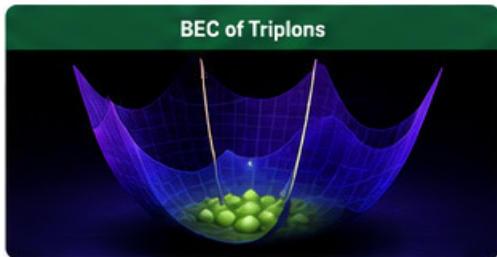
Helium-free sub-Kelvin cooling for quantum technologies



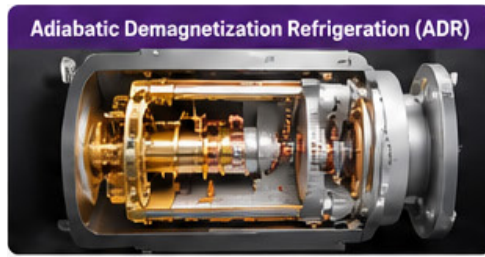
Quantum Spin Liquid (QSL)



Spin Nematicity



BEC of Triplons



Adiabatic Demagnetization Refrigeration (ADR)

Cryogenic Technologies & Research Infrastructure

From Materials to Cryogenic Innovation



Materials Synthesis

Design and synthesize quantum materials with tailored properties

Characterization

Probe quantum phases and correlations using advanced instrumentation

Cryogenic Engineering

Develop ADR systems and probes for extreme low-temperature environments

Research Infrastructure & Instrumentation

Materials Synthesis



- High-temperature furnaces up to 1600 °C
- Split tube furnace with gas-flow and vacuum capability
- Turbo molecular pumping station for ultra-high vacuum

Quantum Characterization



- Physical Property Measurement System (PPMS) / MPMS
- Low temperature Solid State NMR

Cryogenic Development



- ADR setup for sub-Kelvin cooling
- Probe design and development for advanced quantum experiments

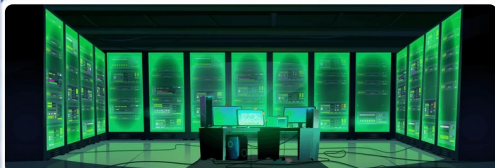


Enabling advanced quantum materials research through integrated synthesis, characterization and cryogenic engineering.

HPC Infrastructure - System Specifications



1. SERVER – COMPUTE NODE HIGH MEMORY



Key Specifications



2 × Intel® Xeon® Platinum 8592V
64 Cores / 128 Threads



24 × 96GB DDR5 5600 ECC REG
(Total 2,304 GB RAM)



2 × 10GBase-T Intel X550, RoHS
10GbE Networking



1 × 960GB SATA 2.5" SSD
High Speed Storage



2 × 1200W Redundant Titanium PSU
(96%) Hot-plug Power Supplies



Ideal For

Quantum Simulations • Large Scale AI Models
Data-Intensive Applications



2. SERVER – COMPUTE NODE LOW MEMORY



Key Specifications



2 × Intel® Xeon® Platinum 8592V
64 Cores / 128 Threads



8 × 64GB DDR5 5600 ECC REG
(Total 512 GB RAM)



2 × 10GBase-T Intel X550, RoHS
10GbE Networking



1 × 960GB SATA 2.5" SSD
High Speed Storage



2 × 1200W Redundant Titanium PSU
(96%) Hot-plug Power Supplies



Ideal For

Parallel Computing • Scientific Workloads
General Purpose Applications



3. SERVER – GPU NODE



Key Specifications



2 × Intel® Xeon® Gold 6530
32 Cores / 64 Threads



4 × NVIDIA H200 NVL 141GB PCIe 5.0 x16
High Performance GPUs



8 × 64GB DDR5 5600 ECC REG
(Total 512 GB RAM)



1 × 7.68TB NVMe PCIe 5.0 3D TLC U.2
Ultra Fast Storage



2 × RJ45 10GbE Intel® X710-AT2
High Speed Networking



4 × 2700W Redundant Titanium PSU
(96%) Hot-plug Power Supplies



Ideal For

Deep Learning • Generative AI • LLM Training
High Performance Computing



128+

CPU Cores Per Node



NVIDIA

4×

NVIDIA H200 NVL GPUs



2.3 TB

DDR5 ECC Memory



Enterprise Grade
HPC Architecture

